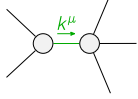
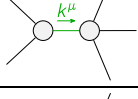
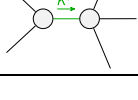
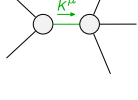
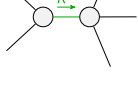
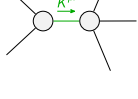


$$\begin{array}{cc}
\mathcal{K}_0^{\#1} \uparrow & \mathcal{K}_0^{\#1} \uparrow^{\alpha\beta\chi} \\
\mathcal{K}_0^{\#1} \uparrow^{\alpha\beta\chi} & \mathcal{K}_0^{\#1} \uparrow^{\alpha\beta\chi}
\end{array}
\begin{array}{cc}
0 & 0 \\
0 & 0
\end{array}
\begin{array}{cccc}
\mathcal{K}_{1^+}^{\#1} \uparrow^{\alpha\beta} & \mathcal{K}_{1^+}^{\#2} \uparrow^{\alpha\beta} & \mathcal{K}_{1^+}^{\#1} \uparrow^{\alpha} & \mathcal{K}_{1^+}^{\#2} \uparrow^{\alpha} \\
\mathcal{K}_{1^+}^{\#1} \uparrow^{\alpha\beta} & -\frac{\gamma_1 k^2}{2} & -\frac{\gamma_1 k^2}{2\sqrt{2}} & 0 & 0 \\
\mathcal{K}_{1^+}^{\#2} \uparrow^{\alpha\beta} & -\frac{\gamma_1 k^2}{2\sqrt{2}} & -\frac{\gamma_1 k^2}{4} & 0 & 0 \\
\mathcal{K}_{1^+}^{\#1} \uparrow^{\alpha} & 0 & 0 & \frac{\gamma_2 k^2}{2} & \frac{\gamma_3 k^2}{2\sqrt{2}} \\
\mathcal{K}_{1^+}^{\#2} \uparrow^{\alpha} & 0 & 0 & \frac{\gamma_3 k^2}{2\sqrt{2}} & \frac{\gamma_4 k^2}{4}
\end{array}
\begin{array}{cc}
\mathcal{K}_{2^+}^{\#1} \uparrow^{\alpha\beta} & \mathcal{K}_{2^+}^{\#1} \uparrow^{\alpha\beta\chi} \\
\mathcal{K}_{2^+}^{\#1} \uparrow^{\alpha\beta\chi} & \mathcal{K}_{2^+}^{\#1} \uparrow^{\alpha\beta\chi}
\end{array}
\begin{array}{cc}
\frac{3\gamma_5 k^2}{2} & 0 \\
0 & \frac{3k^2}{2} \frac{\gamma_{15}^{(4)}}{\gamma_{15}^{(4)}}
\end{array}$$

$$\begin{array}{cc}
\mathcal{T}_0^{\#1} \uparrow & \mathcal{T}_0^{\#1} \uparrow^{\alpha\beta\chi} \\
\mathcal{T}_0^{\#1} \uparrow^{\alpha\beta\chi} & \mathcal{T}_0^{\#1} \uparrow^{\alpha\beta\chi}
\end{array}
\begin{array}{cc}
0 & 0 \\
0 & 0
\end{array}
\begin{array}{cccc}
\mathcal{T}_{1^+}^{\#1} \uparrow^{\alpha\beta} & \mathcal{T}_{1^+}^{\#2} \uparrow^{\alpha\beta} & \mathcal{T}_{1^+}^{\#1} \uparrow^{\alpha} & \mathcal{T}_{1^+}^{\#2} \uparrow^{\alpha} \\
\mathcal{T}_{1^+}^{\#1} \uparrow^{\alpha\beta} & \frac{2}{3\text{Det}(1^+)} & \frac{\sqrt{2}}{3\text{Det}(1^+)} & 0 & 0 \\
\mathcal{T}_{1^+}^{\#2} \uparrow^{\alpha\beta} & \frac{\sqrt{2}}{3\text{Det}(1^+)} & \frac{1}{3\text{Det}(1^+)} & 0 & 0 \\
\mathcal{T}_{1^+}^{\#1} \uparrow^{\alpha} & 0 & 0 & \frac{\gamma_4 k^2}{4\text{Det}(1^+)} & -\frac{\gamma_3 k^2}{2\sqrt{2}\text{Det}(1^+)} \\
\mathcal{T}_{1^+}^{\#2} \uparrow^{\alpha} & 0 & 0 & -\frac{\gamma_3 k^2}{2\sqrt{2}\text{Det}(1^+)} & \frac{\gamma_2 k^2}{2\text{Det}(1^+)}
\end{array}
\begin{array}{cc}
\mathcal{T}_{2^+}^{\#1} \uparrow^{\alpha\beta} & \mathcal{T}_{2^+}^{\#1} \uparrow^{\alpha\beta\chi} \\
\mathcal{T}_{2^+}^{\#1} \uparrow^{\alpha\beta\chi} & \mathcal{T}_{2^+}^{\#1} \uparrow^{\alpha\beta\chi}
\end{array}
\begin{array}{cc}
\frac{1}{\text{Det}(2^+)} & 0 \\
0 & \frac{2}{3k^2} \frac{\gamma_{15}^{(4)}}{\gamma_{15}^{(4)}}
\end{array}$$

Abbreviations used in matrices
$\gamma_1 == 3 \, \kappa_1 - 4 \, \kappa_{15} + 3 \, \kappa_2 + 2 \, \kappa_3$ && $\gamma_2 == 3 \, \kappa_{15} - 2 \, \kappa_2$ && $\gamma_3 == 2 \, \kappa_2 + \kappa_7$ && $\gamma_4 == 3 \, \kappa_1 - 6 \, \kappa_{15} + \kappa_2 - 2 \, \kappa_7$ && $\gamma_5 == \kappa_1 + \kappa_2$ && $\text{Det}(1^+) == -\frac{3}{4} k^2 (3 \, \kappa_1 - 4 \, \kappa_{15} + 3 \, \kappa_2 + 2 \, \kappa_3)$ && $\text{Det}(2^+) == \frac{3}{2} k^2 (\kappa_1 + \kappa_2)$

Lagrangian			
$\begin{aligned} &\overset{(4)}{\kappa_1} \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} + \overset{(4)}{\kappa_2} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} + \overset{(4)}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\alpha\chi} - \\ &3 \, \overset{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} + 6 \, \overset{(4)}{\kappa_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} - 3 \, \overset{(4)}{\kappa_2} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} - \\ &2 \, \overset{(4)}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} - 3 \, \overset{(4)}{\kappa_1} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\chi\alpha} + 3 \, \overset{(4)}{\kappa_{15}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\chi\alpha} - \\ &3 \, \overset{(4)}{\kappa_2} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\chi\alpha} - \overset{(4)}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\chi\alpha} + \overset{(4)}{\kappa_7} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} \partial_\delta \mathcal{K}^{\delta}_{\chi\beta} - \overset{(4)}{\frac{3}{4}} \, \overset{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} - \\ &\overset{(4)}{\frac{3}{4}} \, \overset{(4)}{\kappa_2} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} - \overset{(4)}{\frac{1}{2}} \, \overset{(4)}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} + \overset{(4)}{\kappa_{15}} \partial_\delta \mathcal{K}^{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{\kappa_{15}} \partial_\delta \mathcal{K}^{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$			
Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{T}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{T}_0^{\#1\alpha\beta} == 0$	1	$\begin{aligned} &\partial_\delta \partial^\alpha \mathcal{T}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{T}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{T}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{T}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{T}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{T}^{\beta\alpha\chi} == \\ &\partial_\delta \partial^\alpha \mathcal{T}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{T}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{T}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{T}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{T}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{T}^{\chi\alpha\beta} \end{aligned}$	
$\mathcal{T}_0^{\#1} == 0$	1	$\partial_\beta \mathcal{T}^\alpha_{\alpha\beta} == 0$	
$\mathcal{T}_1^{\#1\alpha\beta} == \mathcal{T}_1^{\#2\alpha\beta}$	3	$3 \, \partial_\delta \partial_\chi \partial^\alpha \mathcal{T}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{T}^{\beta\alpha\chi} + 2 \, \partial_\delta \partial^\delta \partial_\chi \mathcal{T}^{\chi\alpha\beta} == 3 \, \partial_\delta \partial_\chi \partial^\beta \mathcal{T}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{T}^{\alpha\beta\chi}$	
Total # constraint(s):	5		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{1}{-3 \, \overset{(4)}{\kappa_1} + 4 \, \overset{(4)}{\kappa_{15}} - 3 \, \overset{(4)}{\kappa_2} - 2 \, \overset{(4)}{\kappa_3}}$
	1	0	$\frac{\overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_{15}} + \overset{(4)}{\kappa_2}}{\overset{(4)}{\kappa_1} \, \overset{(4)}{\kappa_{15}} + \overset{(4)}{\kappa_{15}} \, \overset{(4)}{\kappa_2}}$
	1	0	$-\frac{\overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_{15}} + \overset{(4)}{\kappa_2}}{\overset{(4)}{\kappa_1} \, \overset{(4)}{\kappa_{15}} + \overset{(4)}{\kappa_{15}} \, \overset{(4)}{\kappa_2}}$
	1	0	$\begin{aligned} &(96 \, \overset{(4)}{\kappa_{15}}^3 \, \overset{(4)}{\kappa_2}^{(4)} - 168 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)2} + 72 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)3} - 48 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} + 48 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} + \\ &8 \, \overset{(4)}{\kappa_1} \, \overset{(4)}{\kappa_{15}}^{(4)} (2 \, \overset{(4)}{\kappa_{15}}^{(4)} - 3 \, \overset{(4)}{\kappa_2}^{(4)} - \overset{(4)}{\kappa_3}^{(4)}) (6 \, \overset{(4)}{\kappa_{15}}^{(4)} - 6 \, \overset{(4)}{\kappa_2}^{(4)} - \overset{(4)}{\kappa_7}^{(4)}) - 16 \, \overset{(4)}{\kappa_{15}}^{(4)2} \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 12 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)2} \, \overset{(4)}{\kappa_7}^{(4)} + \\ &8 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 12 \, \overset{(4)}{\kappa_1}^{(4)} \, \overset{(4)}{\kappa_{15}}^{(4)} (-6 \, \overset{(4)}{\kappa_{15}}^{(4)} + 6 \, \overset{(4)}{\kappa_2}^{(4)} + \overset{(4)}{\kappa_7}^{(4)}) + \sqrt{(\overset{(4)}{\kappa_{15}}^{(4)2} (3 \, \overset{(4)}{\kappa_1}^{(4)} - 4 \, \overset{(4)}{\kappa_{15}}^{(4)} + 3 \, \overset{(4)}{\kappa_2}^{(4)} + 2 \, \overset{(4)}{\kappa_3}^{(4)})^2} \\ &(9 \, \overset{(4)}{\kappa_1}^{(4)4} + 324 \, \overset{(4)}{\kappa_{15}}^{(4)4} + 1161 \, \overset{(4)}{\kappa_2}^{(4)4} + 372 \, \overset{(4)}{\kappa_2}^{(4)3} \, \overset{(4)}{\kappa_7}^{(4)} + 30 \, \overset{(4)}{\kappa_2}^{(4)2} \, \overset{(4)}{\kappa_7}^{(4)2} + 4 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)3} + \overset{(4)}{\kappa_7}^{(4)4} - \\ &6 \, \overset{(4)}{\kappa_1}^{(4)3} (9 \, \overset{(4)}{\kappa_{15}}^{(4)} - 6 \, \overset{(4)}{\kappa_2}^{(4)} + 2 \, \overset{(4)}{\kappa_7}^{(4)}) + 108 \, \overset{(4)}{\kappa_{15}}^{(4)3} (\overset{(4)}{\kappa_2}^{(4)} + 2 \, \overset{(4)}{\kappa_7}^{(4)}) + 9 \, \overset{(4)}{\kappa_{15}}^{(4)2} (117 \, \overset{(4)}{\kappa_2}^{(4)} + 12 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 8 \, \overset{(4)}{\kappa_7}^{(4)2}) + \\ &3 \, \overset{(4)}{\kappa_1}^{(4)2} (375 \, \overset{(4)}{\kappa_{15}}^{(4)2} - 810 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} + 402 \, \overset{(4)}{\kappa_2}^{(4)2} - 128 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 116 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 10 \, \overset{(4)}{\kappa_7}^{(4)2}) - \\ &6 \, \overset{(4)}{\kappa_{15}}^{(4)} (387 \, \overset{(4)}{\kappa_2}^{(4)3} + 68 \, \overset{(4)}{\kappa_2}^{(4)2} \, \overset{(4)}{\kappa_7}^{(4)} - 5 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)2} - 2 \, \overset{(4)}{\kappa_7}^{(4)3}) + 2 \, \overset{(4)}{\kappa_1}^{(4)} (162 \, \overset{(4)}{\kappa_{15}}^{(4)3} + 9 \, \overset{(4)}{\kappa_{15}}^{(4)2} (119 \, \overset{(4)}{\kappa_2}^{(4)} + 10 \, \overset{(4)}{\kappa_7}^{(4)}) - \\ &3 \, \overset{(4)}{\kappa_{15}}^{(4)} (783 \, \overset{(4)}{\kappa_2}^{(4)2} + 132 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} - 7 \, \overset{(4)}{\kappa_7}^{(4)2}) + 2 (585 \, \overset{(4)}{\kappa_2}^{(4)3} + 183 \, \overset{(4)}{\kappa_2}^{(4)2} \, \overset{(4)}{\kappa_7}^{(4)} + 15 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)2} + \overset{(4)}{\kappa_7}^{(4)3})))) / \\ &(\overset{(4)}{\kappa_{15}}^{(4)} (\overset{(4)}{\kappa_1}^{(4)} + \overset{(4)}{\kappa_2}^{(4)}) (-3 \, \overset{(4)}{\kappa_1}^{(4)} + 4 \, \overset{(4)}{\kappa_{15}}^{(4)} - 3 \, \overset{(4)}{\kappa_2}^{(4)} - 2 \, \overset{(4)}{\kappa_3}^{(4)}) (-9 \, \overset{(4)}{\kappa_1}^{(4)} \, \overset{(4)}{\kappa_{15}}^{(4)} + 18 \, \overset{(4)}{\kappa_{15}}^{(4)2} + 6 \, \overset{(4)}{\kappa_1}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} - \\ &15 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} + 6 \, \overset{(4)}{\kappa_2}^{(4)2} + 6 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + \overset{(4)}{\kappa_7}^{(4)2})) \end{aligned}$
	1	0	$\begin{aligned} &(96 \, \overset{(4)}{\kappa_{15}}^3 \, \overset{(4)}{\kappa_2}^{(4)} - 168 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)2} + 72 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)3} - 48 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} + 48 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} + \\ &8 \, \overset{(4)}{\kappa_1} \, \overset{(4)}{\kappa_{15}}^{(4)} (2 \, \overset{(4)}{\kappa_{15}}^{(4)} - 3 \, \overset{(4)}{\kappa_2}^{(4)} - \overset{(4)}{\kappa_3}^{(4)}) (6 \, \overset{(4)}{\kappa_{15}}^{(4)} - 6 \, \overset{(4)}{\kappa_2}^{(4)} - \overset{(4)}{\kappa_7}^{(4)}) - 16 \, \overset{(4)}{\kappa_{15}}^{(4)2} \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 12 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)2} \, \overset{(4)}{\kappa_7}^{(4)} + \\ &8 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 12 \, \overset{(4)}{\kappa_1}^{(4)} \, \overset{(4)}{\kappa_{15}}^{(4)} (-6 \, \overset{(4)}{\kappa_{15}}^{(4)} + 6 \, \overset{(4)}{\kappa_2}^{(4)} + \overset{(4)}{\kappa_7}^{(4)}) - \sqrt{(\overset{(4)}{\kappa_{15}}^{(4)2} (3 \, \overset{(4)}{\kappa_1}^{(4)} - 4 \, \overset{(4)}{\kappa_{15}}^{(4)} + 3 \, \overset{(4)}{\kappa_2}^{(4)} + 2 \, \overset{(4)}{\kappa_3}^{(4)})^2} \\ &(9 \, \overset{(4)}{\kappa_1}^{(4)4} + 324 \, \overset{(4)}{\kappa_{15}}^{(4)4} + 1161 \, \overset{(4)}{\kappa_2}^{(4)4} + 372 \, \overset{(4)}{\kappa_2}^{(4)3} \, \overset{(4)}{\kappa_7}^{(4)} + 30 \, \overset{(4)}{\kappa_2}^{(4)2} \, \overset{(4)}{\kappa_7}^{(4)2} + 4 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)3} + \overset{(4)}{\kappa_7}^{(4)4} - \\ &6 \, \overset{(4)}{\kappa_1}^{(4)3} (9 \, \overset{(4)}{\kappa_{15}}^{(4)} - 6 \, \overset{(4)}{\kappa_2}^{(4)} + 2 \, \overset{(4)}{\kappa_7}^{(4)}) + 108 \, \overset{(4)}{\kappa_{15}}^{(4)3} (\overset{(4)}{\kappa_2}^{(4)} + 2 \, \overset{(4)}{\kappa_7}^{(4)}) + 9 \, \overset{(4)}{\kappa_{15}}^{(4)2} (117 \, \overset{(4)}{\kappa_2}^{(4)} + 12 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 8 \, \overset{(4)}{\kappa_7}^{(4)2}) + \\ &3 \, \overset{(4)}{\kappa_1}^{(4)2} (375 \, \overset{(4)}{\kappa_{15}}^{(4)2} - 810 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} + 402 \, \overset{(4)}{\kappa_2}^{(4)2} - 128 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 116 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 10 \, \overset{(4)}{\kappa_7}^{(4)2}) - \\ &6 \, \overset{(4)}{\kappa_{15}}^{(4)} (387 \, \overset{(4)}{\kappa_2}^{(4)3} + 68 \, \overset{(4)}{\kappa_2}^{(4)2} \, \overset{(4)}{\kappa_7}^{(4)} - 5 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)2} - 2 \, \overset{(4)}{\kappa_7}^{(4)3}) + 2 \, \overset{(4)}{\kappa_1}^{(4)} (162 \, \overset{(4)}{\kappa_{15}}^{(4)3} + 9 \, \overset{(4)}{\kappa_{15}}^{(4)2} (119 \, \overset{(4)}{\kappa_2}^{(4)} + 10 \, \overset{(4)}{\kappa_7}^{(4)}) - \\ &3 \, \overset{(4)}{\kappa_{15}}^{(4)} (783 \, \overset{(4)}{\kappa_2}^{(4)2} + 132 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} - 7 \, \overset{(4)}{\kappa_7}^{(4)2}) + 2 (585 \, \overset{(4)}{\kappa_2}^{(4)3} + 183 \, \overset{(4)}{\kappa_2}^{(4)2} \, \overset{(4)}{\kappa_7}^{(4)} + 15 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)2} + \overset{(4)}{\kappa_7}^{(4)3})))) / \\ &(\overset{(4)}{\kappa_{15}}^{(4)} (\overset{(4)}{\kappa_1}^{(4)} + \overset{(4)}{\kappa_2}^{(4)}) (-3 \, \overset{(4)}{\kappa_1}^{(4)} + 4 \, \overset{(4)}{\kappa_{15}}^{(4)} - 3 \, \overset{(4)}{\kappa_2}^{(4)} - 2 \, \overset{(4)}{\kappa_3}^{(4)}) (-9 \, \overset{(4)}{\kappa_1}^{(4)} \, \overset{(4)}{\kappa_{15}}^{(4)} + 18 \, \overset{(4)}{\kappa_{15}}^{(4)2} + 6 \, \overset{(4)}{\kappa_1}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} - \\ &15 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} + 6 \, \overset{(4)}{\kappa_2}^{(4)2} + 6 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + \overset{(4)}{\kappa_7}^{(4)2})) \end{aligned}$
	2	0	$\begin{aligned} &(144 \, \overset{(4)}{\kappa_{15}}^4 - 36 \, \overset{(4)}{\kappa_1}^{(4)} (3 \, \overset{(4)}{\kappa_{15}}^{(4)} - \overset{(4)}{\kappa_2}^{(4)}) - 336 \, \overset{(4)}{\kappa_{15}}^{(4)3} \, \overset{(4)}{\kappa_2}^{(4)} + 339 \, \overset{(4)}{\kappa_{15}}^{(4)2} \, \overset{(4)}{\kappa_2}^{(4)2} - 174 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)3} + 36 \, \overset{(4)}{\kappa_2}^{(4)4} - 72 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} + \\ &114 \, \overset{(4)}{\kappa_{15}}^{(4)2} \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} - 84 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} + 24 \, \overset{(4)}{\kappa_2}^{(4)3} \, \overset{(4)}{\kappa_3}^{(4)} + 48 \, \overset{(4)}{\kappa_{15}}^{(4)3} \, \overset{(4)}{\kappa_7}^{(4)} - 84 \, \overset{(4)}{\kappa_{15}}^{(4)2} \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 36 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)2} \, \overset{(4)}{\kappa_7}^{(4)} - \\ &24 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 24 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 8 \, \overset{(4)}{\kappa_{15}}^{(4)2} \, \overset{(4)}{\kappa_7}^{(4)2} - 14 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 6 \, \overset{(4)}{\kappa_2}^{(4)2} \, \overset{(4)}{\kappa_7}^{(4)2} - 4 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)2} + \\ &4 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)2} + 2 \, \overset{(4)}{\kappa_1}^{(4)} (-144 \, \overset{(4)}{\kappa_{15}}^{(4)3} + 54 \, \overset{(4)}{\kappa_2}^{(4)3} + 24 \, \overset{(4)}{\kappa_2}^{(4)2} \, \overset{(4)}{\kappa_3}^{(4)} + \overset{(4)}{\kappa_{15}}^{(4)2} (309 \, \overset{(4)}{\kappa_2}^{(4)} + 45 \, \overset{(4)}{\kappa_3}^{(4)} - 42 \, \overset{(4)}{\kappa_7}^{(4)}) + \\ &6 \, \overset{(4)}{\kappa_2}^{(4)2} \, \overset{(4)}{\kappa_7}^{(4)2} + 2 \, \overset{(4)}{\kappa_3}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)2} + \overset{(4)}{\kappa_{15}}^{(4)} (-228 \, \overset{(4)}{\kappa_2}^{(4)2} - 78 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} + 36 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} + 12 \, \overset{(4)}{\kappa_3}^{(4)} \, \overset{(4)}{\kappa_7}^{(4)} - 7 \, \overset{(4)}{\kappa_7}^{(4)2})) + \\ &3 \, \overset{(4)}{\kappa_1}^{(4)2} (93 \, \overset{(4)}{\kappa_{15}}^{(4)2} - 2 \, \overset{(4)}{\kappa_{15}}^{(4)} (65 \, \overset{(4)}{\kappa_2}^{(4)} + 12 \, \overset{(4)}{\kappa_3}^{(4)} - 6 \, \overset{(4)}{\kappa_7}^{(4)}) + 2 (18 \, \overset{(4)}{\kappa_2}^{(4)} + 4 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_3}^{(4)} + \overset{(4)}{\kappa_7}^{(4)})) + \\ &\sqrt{((3 \, \overset{(4)}{\kappa_1}^{(4)} - 4 \, \overset{(4)}{\kappa_{15}}^{(4)} + 3 \, \overset{(4)}{\kappa_2}^{(4)} + 2 \, \overset{(4)}{\kappa_3}^{(4)})^2} \\ &(1296 \, \overset{(4)}{\kappa_{15}}^{(4)6} + 9 \, \overset{(4)}{\kappa_1}^{(4)4} (9 \, \overset{(4)}{\kappa_{15}}^{(4)2} - 6 \, \overset{(4)}{\kappa_{15}}^{(4)} \, \overset{(4)}{\kappa_2}^{(4)} + 16 \, \overset{(4)}{\kappa_2}^{(4)2}) - 216 \, \overset{(4)}{\kappa_{15}}^{(4)5} (\overset{(4)}{\kappa_2}^{(4)} - 4 \, \overset{(4)}{\kappa_7}^{(4)}) + 4 \, \overset{(4)}{\kappa_2}^{(4)2} (6 \, \overset{(4)}{\kappa_2}^{(4)2} + \overset{(4)}{\kappa_7}^{(4)2})^2 + \\ &9 \, \overset{(4)}{\kappa_{15}}^{(4)4} (121 \, \overset{(4)}{\kappa_2}^{(4)2} + 40 \, \overset{(4)}{\kappa_2}^{(4)} \, \overset{(4)}{\kappa_7}^{($